

MOHSEN RAHIMI

Ph.D. Candidate in Economics · University of Bergamo, Italy
mohsen.rahimi [at] unibg [dot] it · mohsenrahimi.com

RESEARCH INTERESTS

Political Economy, NLP and Large Language Models, Explainable AI, Machine Learning for Social Science, Applied Microeconomics.

EDUCATION

- 2024 – present **Ph.D. in Economics**, University of Bergamo
Advisor: Prof. Sergio Galletta (ETH Zürich / Sapienza University of Rome)
- 2020 – 2023 **M.Sc. in Economics and Data Analysis** (Data Science Curriculum), University of Bergamo
Thesis: "Enhancing Trust in AI: An Exploration of Interpretability Systems in NLP Models"
- 2012 – 2018 **B.Sc. in Rock Mechanical Engineering**, University of Tehran

PUBLICATIONS

- [1] Rahimi, M., De Poli, G., Masella, A., & Bregonzio, M. *"Why Did You Fail? An Interpretability System for NLP Models."* Proceedings of ICDSSST 2023, Datrix S.p.A., Milan, Italy.

WORKING PAPERS

- [1] *"ML, Audit Efficiency, and Regional Fairness."* Building on Ash and Galletta's machine learning approach to anti-corruption policy, I analyze how using ML to predict corruption for public audits in Brazil creates unintended regional disparities. I introduce a fairness-constrained optimization step that maintains high corruption detection rates while correcting unequal geographic exposure. Work in progress.
- [2] *"Talk That Matters: Measuring Deliberation Quality in FOMC Transcripts."* Using large language models to analyze Federal Open Market Committee transcripts, moving beyond basic sentiment to measure deliberation features — including speaker responsiveness and directional convergence — to assess whether meeting quality predicts the accuracy of the Fed's subsequent economic forecasts. Work in progress.

PROFESSIONAL EXPERIENCE

- 2022 – 2023 **Data Scientist**, Datrix AI Solution Group (3rdPlace), Milan
– Developed an Adversarial Machine Learning framework for NLP, combining SHAP-based local explanations with global model insights to reduce bias and interpretation variability
– Created an Explainable AI module (AIA Guard) to the Italian Government's CybersecH project
– Led cloud infrastructure deployment on Kubernetes and Google Cloud for high-performance ML pipelines
- 2021 – 2022 **Scientific Research Assistant**, University of Bergamo
– EDA and data cleaning on ISTAT insurance microdata; Contributed to a customer retention optimization algorithm

TEACHING EXPERIENCE

- 2024 – 2025 **Teaching Assistant**, Machine Learning for Economics (Introduction to R programming language, University of Bergamo)

2016 – 2017 **Teaching Assistant**, Engineering Faculty, Basic of Programming languages, University of Tehran

2012 – 2016 **Teacher**, Salam High School, Tehran (C++, Python, Mathematics)

AWARDS & HONORS

2021 **1st Place, Stat-Hackathon Competition**, University of Bergamo
Fraud detection challenge on UniCredit bank transactions (Capgemini, A2A Smart City, IN.TWIG, OROBIX)

2021 **Top 10 Finalist, Future of Mobility Program** (with Harvard Graduate School of Design)
Ranked 9th of 100+ participants; project on agent-based simulation of disruption behavior

SKILLS

<i>Programming</i>	Python, Stata, R, Java, C/C++
<i>Economics</i>	Econometrics, Policy Modeling, Causal Inference, Behavioral Analysis
<i>ML / AI</i>	PyTorch, TensorFlow/Keras, HuggingFace, SHAP, Scikit-learn, H2O, Tidymodels
<i>Data & Tools</i>	PostgreSQL, Neo4j, Docker, Kubernetes, Google Cloud, Git, Linux, LaTeX
<i>Languages</i>	Persian (native), English (IELTS 7.0, 2025), Italian (CILS B2)

REFERENCES

Sergio Galletta

Associate Professor, Sapienza University of Rome; Senior Researcher, ETH Zürich

Ph.D. Supervisor

sergio.galletta@gess.ethz.ch · <http://sergio-galletta.com>

Vittorio Moriggia

Associate Professor, Computer Science in Finance

Department of Management, Economics and Quantitative Methods, University of Bergamo

vittorio.moriggia@unibg.it